# Prodigy Anesthesia Quick Review

## Diuretics

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| In general, how do diuretics increase urinary output? | Diuretics increase urinary output by reducing the amount of sodium and water reabsorbed by the nephron.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 645.* |
| How do all loop diuretics exert their action? | All loop diuretics work by inhibiting sodium and chloride reabsorption in the thick ascending limb of the loop of Henle.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 646.* |
| What are potential adverse side effects of loop diuretics? | Loop diuretics cause an increase in sodium delivery to the distal tubule and collecting tubule eliciting an increase in potassium and hydrogen ion secretion at these locations. This can result in hypokalemia and metabolic alkalosis. The increased urinary excretion of calcium can result in urinary calculi and possibly hypocalcemia. Hypomagnesemia may also result due to increased magnesium excretion. Reversible hearing loss has also been reported with the use of furosemide and ethacrynic acid.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 647.* |
| How do loop diuretics affect electrolyte balance? | Loop diuretics increase the excretion of sodium, chloride, calcium, and magnesium and can potentially result in symptoms related to low serum levels of these electrolytes in addition to the signs and symptoms of hypovolemia due to increased water excretion.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 646.* |
| Where do carbonic anhydrase inhibiting diuretics exert their effect? | Carbonic anhydrase inhibitors such as acetazolamide decrease sodium reabsorption and H+ secretion in the proximal tubules. They can result in a mild hyperchloremic metabolic acidosis.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 648.* |

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| How do thiazide-type diuretics exert their action? | Thiazide-type diuretics such as thiazides (obviously), chlorthalidone, quinethazone, metolazone, and indapamide act on the distal tubule and connecting segment. The inhibit sodium reabsorption which results in impaired dilutional capability. In contrast to loop diuretics, thiazide-type diuretics enhance the reabsorption of calcium in the distal tubule. Of interest is indapamide which has intrinsic vasodilating capabilities and is the only thiazide-type diuretic that undergoes significant hepatic excretion.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 647.* |
| What are the potentially adverse side effects of thiazide-type diuretics? | Thiazide diuretics can impair renal diluting capacity resulting in hyponatremia. They can also result in hyperglycemia, hyperuricemia, hypercalcemia, and hyperlipidemia.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 647.* |
| Do both thiazide-type diuretics and loop diuretics result in hypokalemia? | Yes. Thiazide diuretics do not inhibit sodium reabsorption to the degree that loop diuretics do, thus the amount of sodium delivered to the collecting tubules is less. This results in less potassium secretion than occurs with loop diuretics, but is still sufficient to produce hypokalemia and metabolic alkalosis.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 647.* |
| How do bumetanide and triamterene differ in their effect on potassium levels? | Bumetanide is a loop diuretic associated with decreased serum potassium levels. Drugs such as triamterene, spironolactone, NSAIDs, ACE inhibitors, and beta blockers are associated with increased potassium levels.<br>*Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 488.* |
| How do the noncompetitive potassium-sparing diuretics exert their action? | Unlike spironolactone, triamterene and amiloride do not rely on aldosterone activity. They inhibit sodium reabsorption and potassium secretion by limiting the number of open sodium channels in the lumen of the collecting tubules.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 648.* |
| What is the chief indication for the administration of the potassium-sparing diuretic spironolactone? | Spironolactone is a direct aldosterone receptor antagonist that acts to inhibit aldosterone-mediated sodium reabsorption and potassium secretion in the collecting tubules. As a result, it is only indicated for patients with hyperaldosteronism.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 648.* |

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| How does the efficacy of potassium-sparing diuretics compare with that of loop diuretics and thiazide diuretics? | Potassium sparing diuretics are relatively weak agents that inhibit sodium reabsorption in the collecting tubules. They excrete about 10% of the amount of sodium (and subsequently water) that loop diuretics do and about 30% of the sodium that thiazide diuretics secrete.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 647.* |
| How do the potential adverse effects of noncompetitive potassium-sparing diuretics differ from that of loop and thiazide diuretics? | Amiloride and triamterene can result in hyperkalemia and metabolic acidosis whereas loop diuretics and thiazide diuretics can result in hypokalemia and metabolic alkalosis.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 648.* |
| What accounts for the resistance to diuretics seen in patients with impaired renal function? | Diuretics are highly protein bound which mean that very little of the drug enters the tubules by filtration. Because diuretics exert their action within the lumen of the tubule, most of them must be secreted into the lumen by the proximal tubule via an organic anion pump. In patients with decreased renal function, this mechanism is not effective, therefore less drug is able to cross into the tubule to exert an effect.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 736.* |
| How do osmotic diuretics such as mannitol increase urinary output? | Osmotic diuretics are filtered at the glomerulus and undergo little to no reabsorption in the proximal tubule. These large molecules exert a strong attraction to water and thereby limit the reabsorption of water that normally follows sodium reabsorption. As a result, water excretion is increased.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 736.* |
| What are the potential adverse effects that could be attributed to the administration of mannitol? | Because the administration of mannitol results in an acute increase in plasma and extracellular osmolality, fluid shifts from the intracellular space into the extracellular space and eventually the intravascular compartment resulting in congestive heart failure. Additionally, the increase in plasma volume results in hemodilution which can present as transient hyponatremia and a drop in hematocrit.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 737.* |
| What results would you expect from the administration of mannitol in patients with hypovolemia or severe, renal parenchymal injury? | In patients with hypovolemia, mannitol will augment urinary output. In patients with severe glomerular or tubular injury, however, mannitol will have little effect on urinary output.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 736.* |

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| A patient exhibits hyponatremia with a high serum osmolarity. What diuretic would most likely cause this phenomenon? | Hyponatremia with a normal or high serum osmolarity result from the presence of non-sodium solutes such as mannitol.<br><br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 342.* |
| What diuretic would be most appropriate for a patient allergic to sulfonamide drugs? | The only diuretic (other than osmotic diuretics such as mannitol) that is not derived from sulfonamides is ethacrynic acid which may make it the drug least likely to result in an allergic reaction in patient with sulfa allergies.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: .* |

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